

Please put your answers in a PDF file; name it if-then-else.pdf.
Other file formats will NOT be taken.

Both text and diagrams are involved. MS Office, OpenOffice, or LibreOffice can do great for mixing text and drawings.

Question 1: 3 marks
2: 2 marks
3: 1 mark

Question 1

Most programming languages support if-then-else statements, with the "else" being optional. A long time ago, this was done carelessly and resulted in ambiguous grammars. Here is a simplified re-living of that time, with test conditions and statements replaced by terminal symbols to show the gist.

(Start symbol: <stmt>)
<stmt> ::= <cond> | "A" | "B" | "C"
<cond> ::= "if" <test> "then" <stmt> ["else" <stmt>]
<test> ::= "T1" | "T2"

Draw two different parse trees for

if T1 then if T2 then A else B

Reminder: A parse tree, rather than an abstract syntax tree, is required.

```
1:
<stmt>
|
<cond>
|-- "if"
|-- <test> ("T1")
|-- "then"
|-- <stmt>
|
|   <cond>
|   |-- "if"
|   |-- <test> ("T2")
|   |-- "then"
|   |-- <stmt> ("A")
|   |-- "else"
|   |-- <stmt> ("B")
```

```
2:
<stmt>
|
<cond>
|-- "if"
|-- <test> ("T1")
|-- "then"
|-- <stmt>
|
|   <cond>
|   |-- "if"
|   |-- <test> ("T2")
|   |-- "then"
|   |-- <stmt> ("A")
|-- "else"
|-- <stmt> ("B")
```

Question 2

A way out is to add brackets or equivalent to mark endings (even beginnings), which is adopted by many languages. Here is how Bourne shell does it:

```
<stmt> ::= <cond> | "A" | "B" | "C"
<cond> ::= "if" <test> "then" <stmt> [ else <stmt> ] "fi"
<test> ::= "T1" | "T2"
```

(For simplicity, I omit "elif".)

Add "fi"s to

```
    if T1 then if T2 then A else B
```

to fit this grammar. There are two versions, inspired by the two parse trees witnessed in Question 1; give both versions. (No need to draw the new parse trees.)

1: if T1 then if T2 then A else B fi fi

2: if T1 then if T2 then A fi else B fi

Question 3

But if "else" is compulsory, there is suddenly no ambiguity! This is a glimpse of how CFG ambiguity is very tricky.

```
<stmt> ::= <cond> | "A" | "B" | "C"
<cond> ::= "if" <test> "then" <stmt> "else" <stmt>
<test> ::= "T1" | "T2"
```

Draw a parse tree for

```
    if T1 then if T2 then A else B else C
```

(and discover that you have only one choice).

```
<stmt>
|
<cond>
|-- "if"
|-- <test> ("T1")
|-- "then"
|-- <stmt>
|
|   <cond>
|   |-- "if"
|   |-- <test> ("T2")
|   |-- "then"
|   |-- <stmt> ("A")
|   |-- "else"
|   |-- <stmt> ("B")
|-- "else"
|-- <stmt> ("C")
```